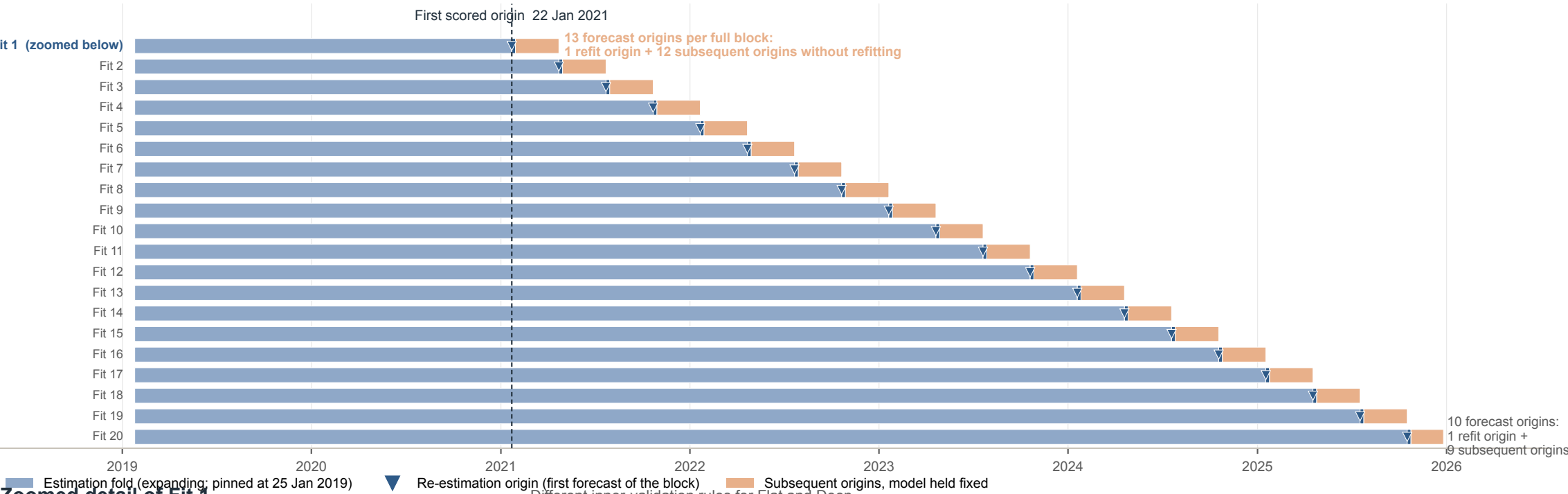


Expanding-window walk-forward forecasting protocol

365 Fridays - 3 leading weeks - 1 trailing week = 361 eligible sequences = 104 warm-up + 257 OOS origins

Common outer calendar with expanding estimation folds and periodic re-estimation



Flat — inner validation, then full-fold refit

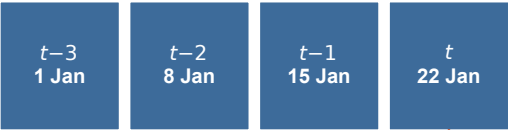


Ridge: 5 α values; XGBoost: 8 configurations · Fit 2: first 65 weeks + last 52 weeks

Select best hyperparameters

Refit on all 104 weeks

Fitted model at origin t



4-week input · P_t known at t

$$\hat{r}_{t+1}$$

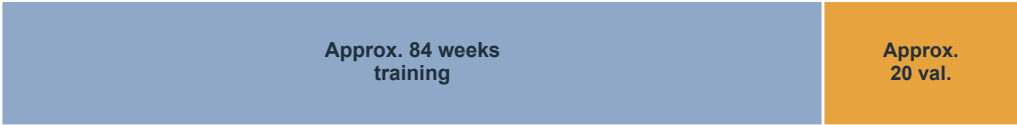
$$\hat{P}_{t+1|t} = P_t \exp(\hat{r}_{t+1})$$

compare with realised P_{t+1}

M0: $P_t \rightarrow \hat{P}_{t+1|t} = P_t$

Accumulate 257 price forecast errors
→ Price RMSE
→ RMSE improvement vs M0 (%)

Deep — dynamic trailing validation, early stopping



Validation length $\approx \min(52, 20\% \text{ of the current estimation fold})$ · Fit 2: approx. 94 train + 23 validation

Early stopping, patience = 12

Retain best checkpoint
No full-fold refit

The estimation fold expands from an initial 104 sequences and is re-estimated every 13 forecast origins. Flat models use the final 52 training sequences for hyperparameter validation and are subsequently refit on the full estimation fold. Deep models use a dynamically sized trailing validation segment for early stopping and retain the best checkpoint without full-fold retraining. Both model families are evaluated on the same 257 out-of-sample forecast origins.